

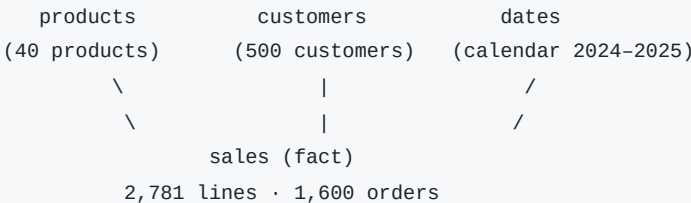
StyleHub Sales Dashboard — Power BI Project

An end-to-end Power BI dashboard for **StyleHub**, a fictional online clothing retailer. Built from scratch: synthetic data → star schema → DAX → three report pages → business recommendations.

The business questions

1. **Are we growing?** Revenue trend, 2024 vs 2025
2. **What actually makes money?** Categories by profit and margin, not just revenue
3. **Who buys?** Customer segments, regions, top customers
4. **What should we do next?** Stock, pricing, and retention actions

Data model (star schema)



Three relationships, single cross-filter direction, dimensions on the 1 side:

- `products[ProductID]` (1) → `sales[ProductID]` (*)
- `customers[CustomerID]` (1) → `sales[CustomerID]` (*)
- `dates[Date]` (1) → `sales[OrderDate]` (*)

Revenue is **not stored in the data** — it's calculated in DAX as `Quantity × UnitPrice × (1 - Discount)`. Deliberate: it forces the measures-vs-calculated-columns distinction that interviewers ask about.

DAX measures (`dax/measures.dax`)

Measure	Definition
Total Revenue	<code>SUMX(sales, Quantity × UnitPrice × (1 - Discount))</code> via <code>RELATED(products[UnitPrice])</code>
Total Cost	<code>SUMX(sales, Quantity × RELATED(products[UnitCost]))</code>

Measure	Definition
Total Profit	[Total Revenue] - [Total Cost]
Profit Margin	DIVIDE([Total Profit], [Total Revenue])
Total Orders	DISTINCTCOUNT(sales[OrderID])
Total Units	SUM(sales[Quantity])
Avg Order Value	DIVIDE([Total Revenue], [Total Orders])

Dashboard pages

1. **Overview** — KPI cards (Revenue, Profit, Orders, AOV), monthly revenue trend split by year, profit by category, revenue by region, Year + Category slicers with full cross-filtering
2. **Products** — Category × Product matrix (revenue, profit, margin, units), Top-10 products bar chart (Top-N filter)
3. **Customers** — revenue by segment, customer ranking table

Key findings

- **\$308,734 revenue · \$170,292 profit · 55.2% margin** across 1,600 orders (2,781 line items). Avg order value **\$192.96**.
- **2025 is outpacing 2024:** 158.78K vs 149.95K revenue, 860 vs 740 orders. Holiday spike every Nov–Dec — seasonality is the strongest pattern in the data.
- **Jackets dominate:** 120.1K revenue (39% of total), 66.6K profit, and the top 3 products overall (Leather Jacket, Puffer Coat, Parka) are all jackets.
- **But margin tells a different story:** Blouses carry the highest category margin at **58.9%**; Pants the lowest at 50.3%. Revenue leaders ≠/margin leaders.
- **Returning customers are the engine:** ~140K revenue vs ~57K from new customers. Retention beats acquisition here.
- **South is the top region** at 29.5% of revenue.
- **Top customer: Stella Garcia** — 3,418.45 across 11 orders.

Data-quality catch

The customer ranking groups by `CustomerName` — but names aren't unique. There are **two different "Mia Taylor"s** (C-0172 in the West, C-0430 in the North) whose spending merges into a single row of ~\$6,335, which would wrongly crown her the #1 customer. Lesson: always key

analysis on `CustomerID`, never on names. (This is the kind of thing that impresses in interviews — you found it, you can explain the fix.)

Recommendations

1. **Protect jacket inventory before Q4** — stock out risk on the Leather Jacket / Puffer Coat / Parka going into the Nov–Dec spike.
2. **Push Blouses harder** — highest margin category (58.9%) deserves more marketing weight relative to its 36.7K revenue.
3. **Retention program for Returning/VIP** — they drive ~80% of revenue; a small lift in repeat purchase rate beats new-customer acquisition spend.
4. **Investigate Pants margin (50.3%)** — lowest in the catalog; check whether discounting or unit costs are the drag.

What's in this repo

```
data/
  sales.csv      # 2,781 order lines, Jan 2024 – Dec 2025 (fact table)
  products.csv   # 40 products, 5 categories (dimension)
  customers.csv  # 500 customers, segments + regions (dimension)
  dates.csv      # calendar table for time intelligence
dax/
  measures.dax   # all 7 measures, copy-paste ready
guide/
  build-guide.md # full step-by-step: install → finished dashboard
screenshots/    # add your page exports here
```

How to open it

1. Open Power BI Desktop → **Get data** → **Text/CSV** → load the four CSVs
2. **Model view**: create the three relationships listed above (dimensions on the `1` side, single filter direction)
3. Add the measures from `dax/measures.dax` (**Home** → **New measure**)
4. Build the three pages, or open the `.pbix` if included

For recruiters

Built to demonstrate: star-schema modeling, DAX (row context with `SUMX` + `RELATED`, `DIVIDE` for safe division), Top-N filtering, slicer cross-filtering, date hierarchies done right (MonthName sorted by MonthNumber), and — most importantly — turning a dashboard into recommended actions.